

Lab 1 Essential Libraries and Tools

• NumPy

EN tutorial: https://www.w3schools.com/python/numpy/numpy_intro.asp

CN tutorial: <https://www.runoob.com/numpy/numpy-tutorial.html>

NumPy is a Python library used for working with arrays.

It also has functions for working in domain of linear algebra, fourier transform, and matrices.

NumPy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely.

NumPy stands for **Numerical Python**.

Task 1: In the code cell below, complete the function `insertSecond` which takes two arguments, array `a` and element `b`, and returns an array which `b` is inserted before the second element in `a`.

```
# You should return your result.
import numpy as np
def insertSecond(a, b):
    # YOUR CODE HERE
    x1, x2 = np.split(a,[1])
    x = np.concatenate([x1, np.array([b])])
    return (np.concatenate([x, x2]))
    # Another solution
    # return np.insert(a, 1, [b])
# Test cases
assert np.array_equal(insertSecond(np.array([-5,-10,-12,-6]),5), np.array([-5, 5, -10,
-12, -6]))
assert np.array_equal(insertSecond(np.array([1,2,3]),7), np.array([1, 7, 2, 3]))
assert np.array_equal(insertSecond(np.array([-5,-10,-12,-6]),8), np.array([-5, 8, -10,
-12, -6]))
assert np.array_equal(insertSecond(np.array([1,2,3]),12), np.array([1, 12, 2, 3]))
```

Task 2: In the code cell below, complete the function `mergeArrays` takes two array arguments `a` and `b`. The two arrays are merged, in which process the duplicate elements are removed. Finally the merged array are sorted.

```
# You should return your result.
import numpy as np
def mergeArrays(a,b):
    # YOUR CODE HERE
    x = np.concatenate([a,b])
    x = np.unique(x)
    x = np.sort(x)
    return x

# Test cases
assert np.array_equal(mergeArrays(np.array([1,1,4,8,1]), np.array([2, 3])),
np.array([1, 2, 3, 4, 8]))
assert np.array_equal(mergeArrays(np.array([-5,-10,-10,-6]), np.array([-5, 8, -10, -12,
-6])),np.array([-12, -10, -6, -5, 8]) )
assert np.array_equal(mergeArrays(np.array([1,1,6,8,1]), np.array([2, 3])),
np.array([1, 2, 3, 6, 8]))
```

• SciPy

EN tutorial: https://www.w3schools.com/python/scipy/scipy_intro.php

CN tutorial: <https://www.runoob.com/scipy/scipy-tutorial.html>

SciPy is a scientific computation library that uses [NumPy](#) underneath.

SciPy stands for **Scientific Python**.

It provides more utility functions for optimization, stats and signal processing.

Like NumPy, SciPy is open source so we can use it freely.

SciPy was created by NumPy's creator Travis Olliphant.

• matplotlib

EN tutorial: https://www.w3schools.com/python/matplotlib_intro.asp

CN tutorial: <https://www.runoob.com/matplotlib/matplotlib-tutorial.html>

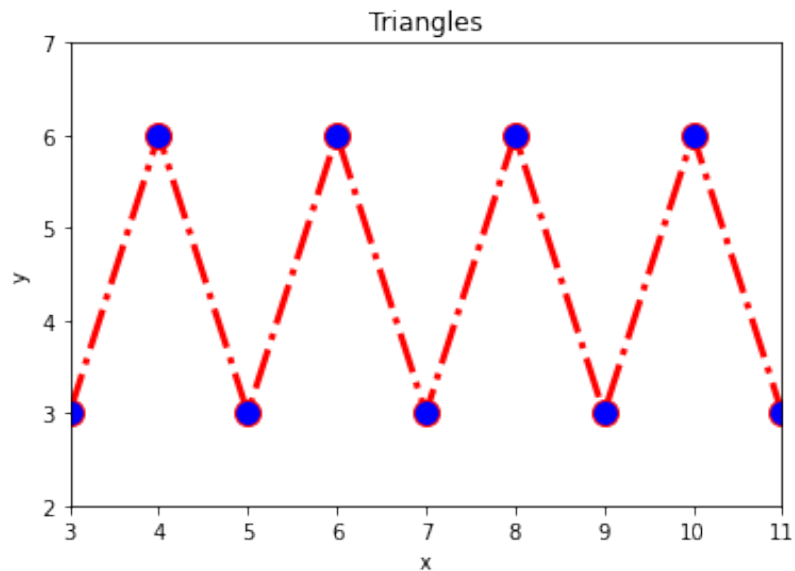
Matplotlib is a low level graph plotting library in python that serves as a visualization utility.

Matplotlib was created by John D. Hunter.

Matplotlib is open source and we can use it freely.

Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

Complete the code below to create a image which looks exactly the same as the follow one:



```
import matplotlib.pyplot as plt

# x axis values
x = [3,4,5,6,7,8,9,10,11]
# y axis values
y = [3,6,3,6,3,6,3,6,3]

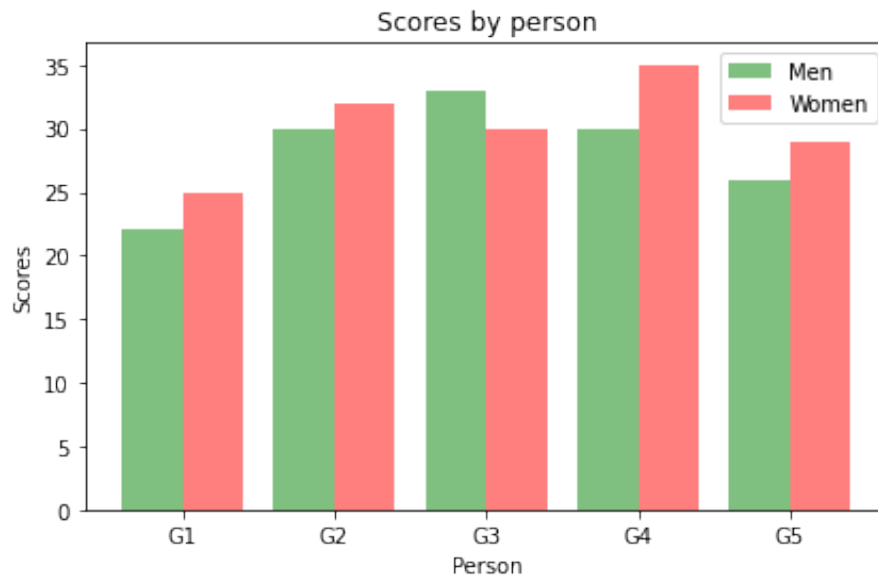
# plotting the points
plt.plot(x, y, color='red', linestyle='dashdot', linewidth = 3,
         marker='o', markerfacecolor='blue', markersize=12)
#Set the y-limits of the current axes.
plt.ylim(2,7)
#Set the x-limits of the current axes.
plt.xlim(3,11)

# naming the x axis
plt.xlabel('x')
# naming the y axis
plt.ylabel('y')

# giving a title to my graph
plt.title('Triangles')

plt.show()
```

Task 3: Complete the code below to create a image which looks exactly the same as the follow one:



```
import numpy as np
import matplotlib.pyplot as plt

# data to plot
n_groups = 5
men_means = (22, 30, 33, 30, 26)
women_means = (25, 32, 30, 35, 29)
alpha = 0.5

# YOUR CODE HERE
### BEGIN SOLUTION
# create plot
fig, ax = plt.subplots()
index = np.arange(n_groups)
bar_width = 0.4
opacity = 0.5

rects1 = plt.bar(index - bar_width/2, men_means, bar_width, alpha=opacity,
color='g',label='Men')
rects2 = plt.bar(index + bar_width/2, women_means, bar_width, alpha=opacity, color='r',
label='Women')

plt.xlabel('Person')
plt.ylabel('Scores')
plt.title('Scores by person')
plt.xticks(index, ('G1', 'G2', 'G3', 'G4', 'G5'))
plt.legend()
plt.tight_layout()
### END SOLUTION

plt.show()
```

• Pandas

EN tutorial: https://www.w3schools.com/python/pandas/pandas_intro.asp

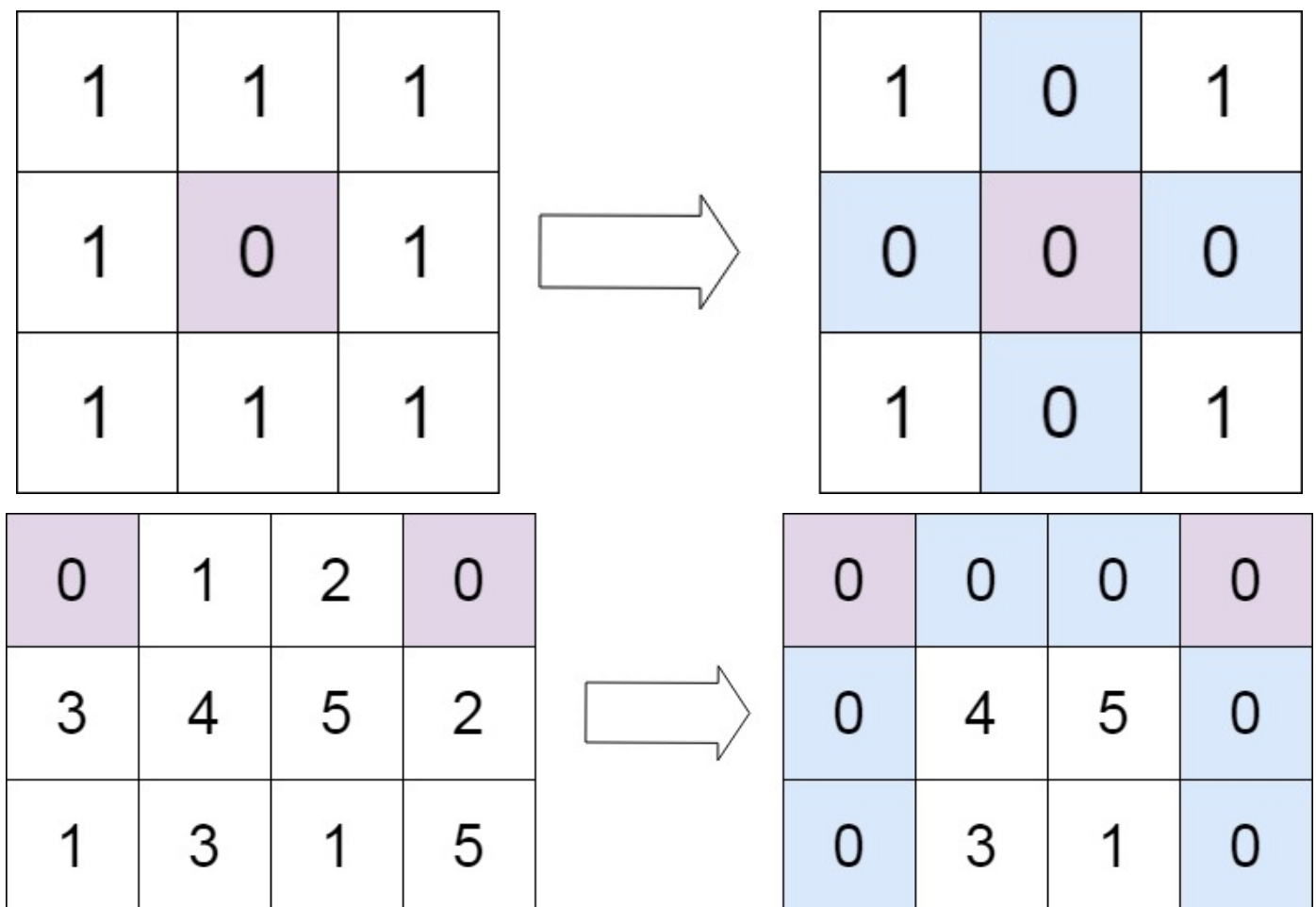
CN tutorial: <https://www.runoob.com/pandas/pandas-tutorial.html>

Pandas is a Python library used for working with data sets.

It has functions for analyzing, cleaning, exploring, and manipulating data.

The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis" and was created by Wes McKinney in 2008.

Task 4: In the code cell below, complete the function `setDataFrameZeros` which takes a dataframe `df` as an argument, and returns a new dataframe. The label and index in the new dataframe is the same as those in `df`. If an element is 0 in `df`, set its entire row and column to 0 in the new dataframe. All other elements are the same as those in `df`.



```
# You should return your result.
import pandas as pd
def setDataFrameZeros(df):
    # YOUR CODE HERE
    ### BEGIN SOLUTION
    matrix = df.values

    rownum = len(matrix)
```

```

colnum = len(matrix[0])
row = [False for i in range(rownum)]
col = [False for i in range(colnum)]
for i in range(rownum):
    for j in range(colnum):
        if matrix[i][j] == 0:
            row[i] = True
            col[j] = True
for i in range(rownum):
    for j in range(colnum):
        if row[i] or col[j]:
            matrix[i][j] = 0

newdata=pd.DataFrame(matrix,index=df.index,columns=df.columns)
return (newdata)
### END SOLUTION

# Test cases
df1 = pd.DataFrame({'c1': [1, 4, 7],
                    'c2': [2, 0, 8],
                    'c3': [3, 6, 9]})
df2 = pd.DataFrame({'c1': [1, 0, 7],
                    'c2': [0, 0, 0],
                    'c3': [3, 0, 9]})
assert (df2.equals(setDataFrameZeros(df1)))

df1 = pd.DataFrame({'c1': [0, 3, 1],
                    'c2': [1, 4, 3],
                    'c3': [2, 5, 1],
                    'c4': [0, 2, 5]})
df2 = pd.DataFrame({'c1': [0, 0, 0],
                    'c2': [0, 4, 3],
                    'c3': [0, 5, 1],
                    'c4': [0, 0, 0]})
assert (df2.equals(setDataFrameZeros(df1)))

df1 = pd.DataFrame({'c1': [1, 4, 7],
                    'c2': [2, 0, 8],
                    'c3': [3, 6, 9]})
df2 = pd.DataFrame({'c1': [1, 0, 7],
                    'c2': [0, 0, 0],
                    'c3': [3, 0, 9]})
assert (df2.equals(setDataFrameZeros(df1)))

df1 = pd.DataFrame({'c1': [0, 3, 1],
                    'c2': [1, 4, 3],
                    'c3': [2, 5, 1],
                    'c4': [0, 2, 5]})
df2 = pd.DataFrame({'c1': [0, 0, 0],

```

```
        'c2': [0, 4, 3],  
        'c3': [0, 5, 1],  
        'c4': [0, 0, 0]})  
assert (df2.equals(setDataFrameZeros(df1)))
```
